InterceptLIVE Demo: Six Rural Critical Care Use Cases

October 23, 2026 · Diego C. Reino, MD

Six scenarios, run on one synthetic Western Nebraska network, show reviewers every NETECH criterion inside the product rather than in the narrative. Each one pairs a chronic-disease exacerbation with a rural constraint, a priority-population patient and a measurable outcome.

Demo environment

The demo runs as one night on a hub-and-spoke network modeled on the Regional West proposal. Every patient, value and note is synthetic; no PHI enters the demo build.

Site	Role	Demo bedside team	Demo EHR	Approx. distance to hub
Regional West, Scottsbluff	Hub; tele-critical-care command center	Hub intensivist, tele-RN	Epic	n/a (Omaha ~450 mi)
Chadron	CAH spoke	1 RN, on-call family physician	Meditech	~100 mi
Alliance	CAH spoke	2 RNs, ED physician	Cerner/Oracle Health	~60 mi
Sidney	Spoke	2 RNs, hospitalist APP	Meditech	~70 mi
Bridgeport	CAH spoke	1 RN, on-call APP	TruBridge	~40 mi
Intercept (remote)	Backup intensivist coverage	Intercept intensivist	n/a	n/a

The EHR assignments are demo assumptions chosen to show multi-EHR normalization; confirm each site's real system in discovery.

Demo rules

- Every synthetic patient carries a payer tag (Medicaid, dual-eligible, uninsured, commercial, Medicare) and a county rurality code. Both feed reporting only and never the AI risk score; the demo should show this setting.
- AI outputs appear as clinician-directed decision support with the inputs shown, never as autonomous orders.
- Each scenario ends by writing its events to the outcomes layer, so Scenario 6 can report on the whole night.
- Build-status tags: **Demo** = must work on synthetic data by Nov 9; **Mock** = clickable screen with static data; **Funded** = built during the award year.

Criteria coverage

Every criterion is shown by at least three scenarios, and each scenario carries one signature capability so the demo never repeats itself.

Scenario	Signature capability	1 Rural	2 Priority consumers	3 Chronic disease	4 Unmet need	5 Quality, cost, access
1 CHF at Chadron, 2 a.m.	AI early warning + keep-or-transfer decision	CAH med-surg bed, 100 mi from hub	Dual-eligible, lives alone	CHF exacerbation	ICU-level care outside an ICU	Transfer avoided, cost logged
2 COPD in a blizzard, Alliance	Low-bandwidth mode + transport-aware decision	Weather, failing network	Uninsured rancher	COPD with hypercapnia	Works when video drops	NIV success, no air transport
3 Sepsis, diabetic CKD, Sidney	Bundle tracker + chronic-aware dosing	ED hold, APP-led	Medicaid expansion adult	Diabetes + CKD	Protocol support for non-intensivists	Bundle compliance, time to antibiotics
4 DKA from insulin rationing, Bridgeport	Nurse-led protocol + SDOH flag	1-RN night	Low-income, cost-driven rationing	Type 1 diabetes	Social cause captured in the ICU	Insulin access referral
5 Surge night, hub view	One-to-many census across 4 EHRs + AI notes	4 sites, 1 intensivist	Payer mix of the whole census	Multiple chronic exacerbations	Hospital-run hub, multi-EHR	Time to intensivist, minutes saved
6 Transition + outcomes	Primary-care handoff, readmission catch, DHHS report	RHC follow-up near home	Payer-stratified outcomes	Readmission prevention	Built-in measurement	All metrics, baseline vs. pilot

Scenario 1: Heart failure at 2 a.m. in a 25-bed hospital

The platform spots a CHF patient decompensating on a med-surg floor three hours before a transfer would normally be called, and an intensivist manages her in place. This is the flagship scenario: it shows criteria 1, 2, 3 and 5 in one story.

Setting. Chadron, 01:40. One RN covers the floor; the family physician is on call from home. No ICU beds. Scottsbluff is about 100 miles away by ground.

Patient (synthetic). Margaret H., 72. HFrEF (EF 25%), stage 3 CKD, type 2 diabetes. Dual-eligible (Medicare + Medicaid). Lives alone 20 miles outside town. Admitted yesterday for a CHF exacerbation on IV furosemide.

Story beats

1. **Early warning.** Overnight, respiratory rate climbs 18 → 26, SpO2 falls to 89% on 4 L, heart rate 112, net fluid balance +1.8 L, weight +2.1 kg over 48 h. The deterioration score crosses threshold at 01:40 and shows the trend and the inputs that drove it.
2. **Network ranking.** On the hub screen Margaret jumps to #1 of all patients across five sites, with a reason line: "rising work of breathing, positive fluid balance, HFrEF."
3. **One-click consult.** The hub intensivist opens video to the bedside RN. Home meds, last echo, creatinine trend and BNP (2,400 pg/mL) are already pulled from the Meditech feed into one panel.
4. **CHF exacerbation pathway.** The pathway suggests BiPAP, escalated IV diuretic with a urine-output target, and nitrate if blood pressure allows. The intensivist accepts, edits one dose, and the order set is sent for local co-signature.
5. **Keep-or-transfer panel.** The panel lists what the plan needs (BiPAP, RT on call, hourly vitals, the RN's capacity) against what Chadron has tonight, plus transport time and the receiving bed. The decision: keep, with checkpoints at 1 h and 4 h that would trigger transfer.
6. **Resolution.** By 06:00 RR is 20 and SpO2 is 95% on 2 L. The platform logs "transfer avoided" with the estimated cost avoided and the travel distance her family didn't have to make.

Capabilities shown

Capability	Status
Deterioration score with explained inputs	Demo
Cross-site risk ranking	Demo
Unified patient panel from a Meditech-style feed	Demo
CHF exacerbation pathway with editable order set	Demo
Keep-or-transfer panel with checkpoints	Demo
Live video from the patient panel	Mock
Live bidirectional Meditech integration	Funded

Metrics written. Time from score threshold to intensivist contact; transfer avoided; estimated transfer cost avoided [$ placeholder until sourced]; ICU-level care hours delivered locally; payer = dual-eligible.

Scenario 2: COPD respiratory failure during a blizzard

The network degrades and transport is grounded, and the platform keeps intensivist guidance flowing anyway. This is the scenario that proves the product was built for rural infrastructure, not adapted to it.

Setting. Alliance ED, 22:15, during a winter storm. Air transport is grounded and the highway to Scottsbluff is closed. The hospital's connection drops to a fraction of normal bandwidth mid-consult.

Patient (synthetic). Dale R., 64, rancher. Severe COPD (GOLD 3), 40 pack-years, home oxygen he often can't afford to refill. Uninsured since leaving a seasonal job. Arterial blood gas on arrival: pH 7.24, PaCO2 78 mmHg.

Story beats

1. **Consult starts on video.** The ED physician starts BiPAP and opens a consult; the hub intensivist reviews the ABG and the chest X-ray.
2. **Network drops.** Video freezes. The platform switches to low-bandwidth mode on its own: audio only, vitals at reduced frequency, and still photos of the BiPAP screen and the next ABG sent as they're taken. A banner shows the mode so nobody thinks data is live when it isn't.
3. **Transport-aware decision.** The keep-or-transfer panel shows air "grounded," ground "road closed, ETA unknown." The plan becomes stabilize here, with an explicit failure plan.
4. **NIV failure tracking.** The COPD pathway sets checkpoints: repeat ABG at 1 h and 2 h, with pH, PaCO2, mental status and work of breathing. The intubation checklist and starting ventilator settings are prepared with the ED physician in advance, in case.
5. **Resolution.** At 2 h, pH is 7.31 and PaCO2 is 64. NIV is succeeding; Dale stays in a monitored bed in Alliance under hub follow-up.
6. **Social needs flagged.** The discharge workflow flags "uninsured" and "can't afford home oxygen," and generates referrals for Medicaid eligibility screening and pulmonary rehab closer to home.

Capabilities shown

Capability	Status
Automatic low-bandwidth fallback with visible mode banner	Demo (simulated network drop)
Store-and-forward images (device screens, ABG)	Demo
Transport and weather status in the keep-or-transfer panel	Mock
COPD pathway with NIV failure checkpoints	Demo
Intubation readiness checklist	Demo
Coverage and social-needs referrals at discharge	Mock
Bandwidth-tested video stack on rural networks	Funded

Metrics written. NIV success without intubation; transfer not required; consult continuity during network degradation (minutes in fallback mode); social-needs referrals made; payer = uninsured.

Clinical review needed. Confirm the ABG values and NIV checkpoint timing against Intercept's COPD protocol.

Scenario 3: Sepsis in a patient with diabetes and kidney disease

An APP running a septic shock resuscitation gets intensivist-level protocol support, with dosing and fluids adjusted for the patient's chronic disease. It shows that the platform raises the quality of care delivered by the local team, not only the remote one.

Setting. Sidney ED, 15:30. The inpatient unit is full, so the patient boards in the ED. A hospitalist APP leads the case with two RNs.

Patient (synthetic). Rosa M., 51. Type 2 diabetes (A1c 10.8%), stage 4 CKD (eGFR 22), hypertension. Medicaid expansion enrollee. Preferred language Spanish. Presents with pyelonephritis: BP 84/48, HR 124, temp 39.4 °C, lactate 3.8 mmol/L, glucose 390 mg/dL.

Story beats

1. **Sepsis screen fires.** The platform flags probable septic shock from vitals and lactate and starts a visible hour-1 bundle timer on both the hub and bedside screens.
2. **Interpreter in the room.** Preferred language triggers a prompt to add a medical interpreter to the video call.
3. **Bundle tracker.** Each element turns green as it's documented: lactate, blood cultures before antibiotics, broad-spectrum antibiotics, fluids, repeat lactate.
4. **Chronic-aware safety checks.** The antibiotic step flags eGFR 22 for renal dose review. The fluid step shows the 30 mL/kg target alongside a fluid-tolerance check, so the APP reassesses after each bolus rather than giving it all blind.
5. **Vasopressor guidance.** MAP stays under 65 after fluids. The intensivist guides peripheral norepinephrine per protocol, with the line-site checks built into the RN task list.
6. **Glucose and resolution.** An insulin infusion protocol starts for glucose control. By 6 h lactate is 1.6; by 12 h pressors are off and Rosa is admitted locally under hub co-management.

Capabilities shown

Capability	Status
Sepsis screen and hour-1 bundle timer on both screens	Demo
Element-by-element bundle compliance tracking	Demo
Renal-dosing and fluid-tolerance prompts driven by chronic conditions	Demo
RN task list generated from intensivist orders	Demo
Interpreter prompt from preferred language	Mock
Live order and documentation write-back	Funded

Metrics written. Time to antibiotics; bundle compliance by element; time to MAP ≥ 65; pressor-free by 24 h; care delivered locally; payer = Medicaid; language = Spanish.

Clinical review needed. Align bundle elements with the measure set you plan to report and with Intercept's peripheral vasopressor protocol.

Scenario 4: DKA caused by insulin rationing, managed by one night nurse

A single RN safely runs a 14-hour insulin-drip protocol with the platform doing the arithmetic and the reminders, and the ICU stay ends by addressing why the patient got sick. It's the clearest demonstration of prevention, the word the fund uses for chronic disease.

Setting. Bridgeport, 23:00. One RN on the inpatient side; an APP on call from home. Hub tele-RN and intensivist cover remotely.

Patient (synthetic). Tyler K., 24, type 1 diabetes since age 11. Works at a grain elevator. Lost Medicaid at redetermination three months ago and has been stretching his insulin. pH 7.08, bicarbonate 6 mEq/L, anion gap 28, glucose 520 mg/dL, potassium 5.6 mEq/L.

Story beats

1. **Protocol launch.** The intensivist starts the DKA pathway from the hub. The RN's screen becomes an hourly checklist: glucose, potassium, gap, fluids, insulin rate.
2. **Built-in calculator.** The insulin rate adjusts from each glucose entry. The protocol blocks insulin if potassium falls below 3.3 and prompts replacement, and it prompts dextrose when glucose falls below 250.
3. **Hourly nudges.** If a check is 10 minutes overdue, the RN and the hub tele-RN both get an alert. The intensivist signs off rate changes from a phone in under a minute.
4. **Gap closure.** The anion gap closes at hour 13. The platform walks the subcutaneous insulin overlap before the drip stops.
5. **The cause, captured.** The admission note records "cost-related insulin rationing" as a structured social-needs field rather than free text.
6. **Prevention at discharge.** Referrals go out for Medicaid re-enrollment, a manufacturer insulin-assistance program and a diabetes educator visit at the local rural health clinic.

Capabilities shown

Capability	Status
Nurse-facing hourly protocol checklist	Demo
Insulin and potassium rules with hard stops	Demo
Overdue-check alerts to RN and hub	Demo
Mobile sign-off for the intensivist	Mock
Structured social-needs fields	Demo
Referral generation at discharge	Mock
Referral delivery to clinics and assistance programs	Funded

Metrics written. Time to gap closure; protocol adherence (checks on time); hypoglycemia and hypokalemia events; social-needs flag captured; referrals made; payer = uninsured (coverage lapsed).

Clinical review needed. Thresholds above are typical DKA protocol values; replace them with Intercept's protocol before the demo.

Scenario 5: Surge night across five hospitals and four EHRs

One hub intensivist handles two emergencies at once across a 12-patient, five-site census, with overflow to Intercept and AI-drafted notes. This is the unmet-need scenario: a rural hospital running its own regional program, which no incumbent platform is built to support.

Setting. Regional West hub, 03:10, in the Year 3 handover model: a Regional West–employed intensivist is on the console, with Intercept as backup. Census: 12 patients across five sites, drawn from Epic, Cerner, Meditech and TruBridge feeds.

Patients in play (synthetic)

- **Walter P., 71, Regional West ICU.** COPD and obesity, on a ventilator; desaturating to 86%.
- **Linda S., 66, Sidney.** Uncontrolled hypertension; atrial fibrillation, off anticoagulation for 4 months because of cost. Medicaid. Sudden left-sided weakness, last known well 02:30, NIHSS 9.
- **Ten other patients**, including Margaret (Scenario 1), stable and ranked lower.

Story beats

1. **One census, four EHRs.** Every patient shows in the same panel layout, with a small badge for the source system. Nothing in the intensivist's workflow changes by site.
2. **Two events at once.** Walter and Linda both escalate within two minutes. The risk ranking puts both at the top and shows why.
3. **Overflow.** The hub intensivist takes Walter. The platform offers Linda's stroke alert to the Intercept backup intensivist with one click, sending the full panel, CT images and NIHSS; Intercept connects in under 5 minutes.
4. **The right transfer, faster.** Linda receives thrombolysis at Sidney under remote direction. Imaging suggests a large-vessel occlusion, so the platform supports the transfer that should happen: it shows receiving centers, builds the transfer packet automatically and logs door-in-door-out time.
5. **Walter stabilized.** Ventilator adjustments and a bronchodilator plan are entered; his score drops back down the list.
6. **Documentation returned.** For each encounter the AI drafts a note from the data and the call; the intensivist edits and signs. A shift counter shows minutes of documentation saved, and the 07:00 handoff report builds itself.

Capabilities shown

Capability	Status
Unified census across four EHR sources, with source badges	Demo (synthetic feeds)
Simultaneous-event ranking with reasons	Demo
Overflow routing to backup coverage	Demo
Transfer packet and receiving-center view	Mock
AI-drafted notes with clinician edit and sign	Demo

Capability	Status
Auto-generated shift handoff report	Demo
Role-based multi-tenancy for hub, spokes and Intercept	Funded
Live integration with all four EHRs	Funded

Metrics written. Time to intensivist for each event; overflow events and response time; door-in-door-out time for the stroke transfer; documentation minutes saved per shift; patients per intensivist; payer mix of the night's census.

Note. This scenario deliberately includes a transfer, so reviewers see the platform improves transfer decisions in both directions rather than only avoiding them.

Scenario 6: After the ICU, a readmission caught early and the outcomes report

Margaret from Scenario 1 goes home with a primary-care handoff, returns 11 days later and is caught before readmission, and the demo closes on the dashboard that reports all six scenarios. It ties the product to measurable quality, affordability and access, and to the fund's quarterly reporting.

Setting. Chadron and its rural health clinic, days 4 to 11 after Scenario 1; then the hub's outcomes dashboard.

Story beats

1. **Primary-care handoff.** At discharge the platform builds a one-page handoff to Margaret's clinic: what happened, medication changes, daily-weight instructions and a 7-day follow-up. It goes to the clinic in a format it can read without a shared EHR.
2. **Readmission risk.** Her score is high (HFrEF, lives alone, recent ICU-level care), so the platform schedules a 72-hour nurse phone check and puts it on the hub tele-RN's list.
3. **The catch.** On day 11 Margaret comes to the Chadron ED with weight up 1.5 kg and new orthopnea. The platform matches her to the earlier episode and alerts the hub. Her diuretic is adjusted in the ED and she goes home without admission.
4. **Outcomes dashboard.** The demo switches to the pilot dashboard, populated with one synthetic month that includes the scenarios above:
   - **Access:** ICU-level coverage hours at spoke sites; median time from alert to intensivist.
   - **Quality:** transfers retained vs. baseline, bundle compliance, ICU-level length of stay, 30-day readmissions, mortality vs. benchmark.
   - **Affordability:** transfer costs avoided and revenue retained by each hospital.
5. **Priority populations.** One toggle stratifies every metric by payer and by county rurality, showing who benefited.
6. **Reporting.** One click exports the quarterly DHHS report and the evidence pack for the next milestone payment.

Capabilities shown

Capability	Status
Primary-care handoff document	Demo
Readmission risk score and follow-up task	Demo
Cross-site patient matching on return	Demo (synthetic)
Outcomes dashboard with baseline comparison	Mock (static synthetic month)
Payer and rurality stratification	Mock
Quarterly report and milestone evidence export	Mock
Live outcome calculation from production data	Funded
Handoff delivery to clinics outside the EHR	Funded

Metrics written. Readmission avoided; follow-up completed within 7 days; every metric above, stratified by payer and rurality.

Caution. Label the dashboard "synthetic data" on screen during the demo. Only the measured cohort results from Intercept's existing service belong in the application as real outcomes.

Demo run order

The full demo runs about 20 minutes as one continuous night; the 8-minute cut keeps Scenarios 1, 5 and 6, which together still touch all five criteria.

1. **Open on the hub census** (1 min): five sites, four EHRs, 12 patients, payer tags visible but marked "reporting only."
2. **Scenario 1, CHF** (4 min): the flagship story, from early warning to transfer avoided.
3. **Scenario 2, COPD** (3 min): trigger the network drop live.
4. **Scenario 3, sepsis** (3 min): bundle timer and chronic-aware safety checks.
5. **Scenario 4, DKA** (2 min): show the RN's screen, then the social-needs capture.
6. **Scenario 5, surge** (3 min): two events, overflow, AI notes.
7. **Scenario 6, outcomes** (4 min): the readmission catch, then close on the dashboard and the DHHS export.

Build checklist for Nov 9

- ☐ Synthetic dataset: 12 patients across 5 sites, with scripted vital-sign trends, payer and rurality tags
- ☐ Synthetic feeds in four EHR formats normalized into one patient panel
- ☐ Deterioration score and cross-site ranking running on the scripted trends
- ☐ Pathways for CHF, COPD, sepsis and DKA, reviewed and signed off by Intercept clinical leadership
- ☐ Keep-or-transfer panel with checkpoints
- ☐ Low-bandwidth mode with a demo toggle to simulate the network drop
- ☐ Overflow routing to backup coverage
- ☐ AI note drafting with edit and sign
- ☐ Primary-care handoff and readmission follow-up task
- ☐ Outcomes dashboard with one static synthetic month and payer stratification
- ☐ "Synthetic data" watermark on every screen
- ☐ Recorded backup video of the full run, in case the live demo fails
- ☐ Capability claims in the application checked against what the demo actually does